EXPERIMENT 6

DCL Commands: GRANT and REVOKE

Aim: To grant and revoke database privileges for a MySQL user.

Software Required: MySQL 8.0 Command Line Client

SQL Program

```
USE SQL_Lab;

CREATE USER 'labuser'@'localhost'
IDENTIFIED BY 'Lab@12345';

GRANT SELECT, INSERT
ON SQL_Lab.*
TO 'labuser'@'localhost';

SHOW GRANTS FOR 'labuser'@'localhost';

REVOKE INSERT
ON SQL_Lab.*
FROM 'labuser'@'localhost';

SHOW GRANTS FOR 'labuser'@'localhost';
```

```
mysql>
mysql> USE SQL_Lab;
Database changed
mysql>
mysql> CREATE USER 'labuser'@'localhost'
    -> IDENTIFIED BY 'Lab@12345';
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> GRANT SELECT, INSERT
    -> ON SQL_Lab.*
    -> TO 'labuser'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SHOW GRANTS FOR 'labuser'@'localhost';
+-----------------------------------------------------------+
| Grants for labuser@localhost                              |
+-----------------------------------------------------------+
| GRANT USAGE ON *.* TO `labuser`@`localhost`                |
| GRANT SELECT, INSERT ON `sql_lab`.* TO `labuser`@`localhost` |
+-----------------------------------------------------------+
2 rows in set (0.00 sec)

mysql>
mysql> REVOKE INSERT
    -> ON SQL_Lab.*
    -> FROM 'labuser'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SHOW GRANTS FOR 'labuser'@'localhost';
+-----------------------------------------------------+
| Grants for labuser@localhost                        |
+-----------------------------------------------------+
| GRANT USAGE ON *.* TO `labuser`@`localhost`          |
| GRANT SELECT ON `sql_lab`.* TO `labuser`@`localhost` |
+-----------------------------------------------------+
2 rows in set (0.00 sec)

mysql>
```

Observation

Execute the SQL statements in MySQL 8.0 Command Line Client. Verify that each statement runs successfully and that the displayed rows match the inserted sample data and query conditions.

Note: Creating users and changing privileges requires an account with suitable administrative permissions. If labuser already exists, skip CREATE USER. Avoid using a real or reused password in a shared submission.

Result

Thus, Experiment 6 (DCL Commands: GRANT and REVOKE) was performed using MySQL 8.0.